DESIGN AND ANALYSIS OF ALGORITHMS

MICRO PROJECT REPORT

OPTIMAL PREFIX ENTROPY CODING FOR INSTRUCTION MEMORY

How can greedy priority-queue tree construction minimize instruction storage while preserving Shannon entropy limits?

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Course: Kalvium

Subject: DESIGN AND ANALYSIS OF ALGORITHMS

Concept: Greedy Huffman Tree Construction, Min-Heap Priority Queues, and Shannon Entropy

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ALLIANCE SCHOOL OF ADVANCED COMPUTING

Alliance University, Bengaluru

October 2026

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OPTIMAL PREFIX ENTROPY CODING FOR INSTRUCTION MEMORY

How can greedy priority-queue tree construction minimize instruction storage while preserving Shannon entropy limits? --- **Submitted by:** Bandla Praharshitha **Course:** Kalvium **Subject:** Design and Analysis of Algorithms (DAA) **Algorithmic Concept:** Greedy Huffman Tree Construction, Min-Heap Priority Queues, and Shannon Entropy Bounds **Submitted to:** Simran Kumari Course Faculty / Evaluator **ALLIANCE SCHOOL OF ADVANCED COMPUTING Alliance University, Bengaluru October 2026** ---

1. Title

Optimal Prefix Entropy Coding for Micro-Architecture Instruction Memory *Understanding greedy algorithmic optimization, deterministic min-heap tree synthesis, and lossless decompression through instruction memory compression.* ---

2. Problem Statement

Modern microprocessors and embedded systems are constrained by the **memory wall**—the growing performance and bandwidth latency gap between on-chip processors and instruction storage (ROM/RAM). In microprogrammed architectures, control store memory consumes substantial silicon die area and static power. Traditional architectures employ fixed-width instruction words (e.g., 12-bit or 16-bit) for every operation. However, empirical analysis shows that a small subset of instructions (`LOAD` , `ADD` , `NOP` , `HALT`) accounts for over 70% to 80% of executed code, creating vast statistical redundancy. How can we design, implement, and analyze an algorithmically optimal, lossless compression pipeline using David Huffman's greedy prefix algorithm to dramatically compress control store footprints while maintaining deterministic $O(N \log N)$ complexity and exact Shannon entropy coding efficiency? ---

3. Introduction: What Is It?

Have you ever noticed that in any program or microcode sequence, certain instructions appear far more frequently than others? While a compiler or instruction set architecture may provide dozens of instructions, everyday computational loops repetitively cycle through simple arithmetic, data loads, and branches. Under a traditional fixed-length binary encoding scheme, each instruction occupies the exact same number of bits—regardless of whether it runs once or a billion times. This wastes valuable hardware memory bandwidth and storage. Variable-length entropy coding addresses this by assigning shorter binary

codewords to frequent instructions and reserving longer codewords for rare instructions. Among entropy coding techniques, **Huffman Coding** is the provably optimal prefix-free binary character code. I chose this project topic because it directly bridges fundamental algorithmic theory learned in DAA—such as greedy choice strategy, binary min-heap priority queues, and asymptotic complexity—to real-world micro-architectural systems. Through this project, I built a complete Python FastAPI backend engine that computes symbol frequency distributions, constructs deterministic Huffman trees, serializes compressed bitstreams, and provides lossless decompression for instruction sequences. ---

4. How Does It Work?

> `[Execution Pipeline Flowchart Diagram Included]`

Step 1: Frequency Profiling

The raw micro-operation or assembly sequence is scanned. An associative frequency table is generated counting the occurrences of each unique instruction symbol (f_i for symbol s_i).

Step 2: Min-Heap Initialization

For each unique symbol, a leaf node with weight f_i is instantiated. All leaf nodes are inserted into a deterministic min-heap priority queue. Deterministic tie-breaking rules (comparing symbol alphanumeric keys upon weight equality) prevent non-deterministic codebook generation.

Step 3: Greedy Huffman Tree Construction

At each of $N - 1$ iterations, the two forest nodes with the lowest frequencies ($node_1$ and $node_2$) are extracted from the min-heap. A new internal parent node is created with frequency $f_{parent} = f_1 + f_2$, assigning $node_1$ as the left child and $node_2$ as the right child. The parent is inserted back into the heap.

Step 4: Prefix-Free Codebook Derivation

Once a single root tree remains, a depth-first traversal assigns binary `0` to left edges and `1` to right edges. Because all instruction symbols reside strictly at the leaf nodes, no codeword is a prefix of any other codeword (the prefix-free property).

Step 5: Bitstream Packing, Lossless Decoding & Validation

The original instruction sequence is transformed into a compact bitstream by concatenating the derived prefix codes. A companion decoder reconstructs the exact original token stream by traversing the prefix

tree bit-by-bit until reaching a leaf node, verifying 100% lossless fidelity ($\text{Decoded} \equiv \text{Original}$). ---

5. Algorithmic Concept: Greedy Choice, Priority Queues & Shannon Entropy

The Greedy Choice Property

An optimization problem exhibits the greedy choice property if a globally optimal solution can be arrived at by making locally optimal choices at each stage. In Huffman coding, the greedy choice is to always combine the two lowest-frequency subtrees. **Greedy Invariant Proof Outline:** Let x and y be the two characters with the lowest frequencies in an alphabet C. There exists an optimal prefix code for C in which the codewords for x and y have the maximum length and differ only in the last bit (i.e., are sibling leaves at maximum depth). By replacing x and y with a composite symbol z of frequency $f(z) = f(x) + f(y)$, the problem reduces to finding an optimal code for $C' = (C \setminus \{x, y\}) \cup \{z\}$. By mathematical induction, repeated greedy extraction yields a tree with minimum weighted external path length: $$WPL = \sum_{i=1}^N f_i \cdot l_i$$

Min-Heap Priority Queue Complexity

Using an array-backed binary min-heap:

- Initial heap building takes $O(N)$ time.
- Each extraction takes $O(\log N)$ time.
- Performing $N - 1$ joins requires $2(N - 1)$ heap extractions and $N - 1$ insertions.
- **Total Time Complexity:**

$$T(N) = O(N \log N)$$

- **Total Space Complexity:** $O(N)$ to store $2N - 1$ tree nodes and codebook mappings.

Shannon Entropy Limit & Coding Efficiency

Claude Shannon's source coding theorem establishes that the absolute lower theoretical bound for the average codeword length L_{avg} of an information source X is its entropy $H(X)$: $$H(X) = -\sum_{i=1}^N p_i \log_2(p_i) \quad \text{bits/symbol}$$ where $p_i = \frac{f_i}{\sum f_j}$. The average codeword length achieved by the Huffman code is: $$L_{avg} = \sum_{i=1}^N p_i \cdot l_i \quad \text{bits/symbol}$$ By Shannon's theorem: $$H(X) \le L_{avg} < H(X) + 1$$ The **Coding Efficiency** (η) is defined as: $$\eta = \left(\frac{H(X)}{L_{avg}} \right) \times 100\%$$ In our empirical tests, high-skew instruction streams achieved coding efficiencies between **95.8% and 100.0%**. ---

6. Factors Affecting Compression & Algorithmic Performance

1. **Symbol Frequency Skewness (Entropy Variance):** If all instructions appear with equal frequency (uniform distribution), Huffman coding achieves virtually no savings ($L_{avg} \approx \log_2 N$). When frequencies are heavily skewed (Zipfian distribution, typical in assembly code), compression ratios peak above $3.0\times$. 2. **Vocabulary Size (N) vs. Program Length (M):** For small sequences ($M < 10$), the serialized tree metadata overhead ($B_{tree} = (2N - 1) + N \cdot W_{symbol}$) constitutes a noticeable percentage. For realistic microprograms ($M \ge 50$), raw payload compression dominates completely. 3. **Deterministic Heap Tie-Breaking:** Without deterministic secondary keys during frequency collisions, min-heap implementations produce divergent tree topologies across runs, breaking reproducibility across distributed compiler toolchains. 4. **Bits-per-Symbol Baseline ($W_{baseline}$):** Comparing variable-length bitstreams against 8-bit, 12-bit, or 16-bit fixed baselines alters the calculated space savings: $$\text{Space Savings \%} = \left(1 - \frac{B_{payload}}{M \times W_{baseline}}\right) \times 100\%$$ ---

7. Real-Life Applications

- **Embedded Microcontroller Control Stores:** Microcontrollers (e.g., ARM Cortex-M, PIC) use compressed ROM lookups to fit complex firmware into tiny on-chip silicon footprints.
- **Variable-Length Instruction Set Architectures (ISAs):** Modern x86-64 and ARM Thumb-2 architectures employ frequency-biased variable-length instruction formats where common instructions occupy 16 bits and rare instructions occupy 32 or 48 bits.
- **Lossless Network Transmission & Serialization:** Used in HTTP/2 (HPACK header compression relies directly on static/dynamic Huffman coding), JPEG, and DEFLATE (ZIP/GZIP).

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8. Article / Source Review

Source 1: David A. Huffman (Original 1952 Proceedings of the IRE)

- **Title:** *A Method for the Construction of Minimum-Redundancy Codes*
- **Reference:** D. A. Huffman, "A Method for the Construction of Minimum-Redundancy Codes," in *Proceedings of the IRE*, vol. 40, no. 9, pp. 1098-1101, Sept. 1952.
- **Review:** This seminal paper introduced the bottom-up greedy tree construction that guarantees minimum average code length among all uniquely decodable codes. Huffman established that sorting probabilities and repeatedly combining the lowest two elements generates an optimal tree. This paper directly guided my implementation of the deterministic min-heap priority queue and recursive code generation.

Source 2: Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein (MIT Press)

- **Title:** *Introduction to Algorithms (CLRS), 4th Edition - Chapter 16: Greedy Algorithms*
- **Reference:** T. H. Cormen et al., *Introduction to Algorithms*, 4th ed. Cambridge, MA: The MIT Press, 2022.
- **Review:** CLRS provides the rigorous mathematical formulation of the greedy choice property and optimal substructure that underpins Huffman's algorithm. The textbook details why the greedy approach succeeds without backtracking and analyzes priority queue operations ($O(N \log N)$). I used their formal proofs to design the unit test assertions validating prefix-free properties and tree balance.

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9. Experimental Results & Verification

Test Execution Summary

- **Python Backend Test Suite:** 36 automated unit tests passed in 2.13 seconds (`test_huffman_daa.py`, `test_api_endpoints.py`, `test_health.py`).
- **End-to-End Pipeline Verification:** 100% pass rate across roundtrip tests.

Sample Program Compression Benchmark (Arithmetic Accumulation)

- **Input Micro-Operations:** `LOAD R0`, `LOAD R1`, `ADD R0, R1`, `STORE R0, 4`, `INC R1`, `ADD R0, R1`, `HALT`.
- **Original 12-Bit Uncompressed Size:** $7 \times 12 = 84\text{ bits}$.
- **Compressed Bitstream:** `110111010010101100` (18 payload bits).
- **Payload Compression Ratio:** $4.67\times$.
- **Payload Space Savings:** 78.57%.
- **Lossless Verification Verdict:** Exact match (0 bit discrepancy, 100% fidelity).

Screenshot of Running DAA Lab

(Captured live from running Next.js application at `/daa`)

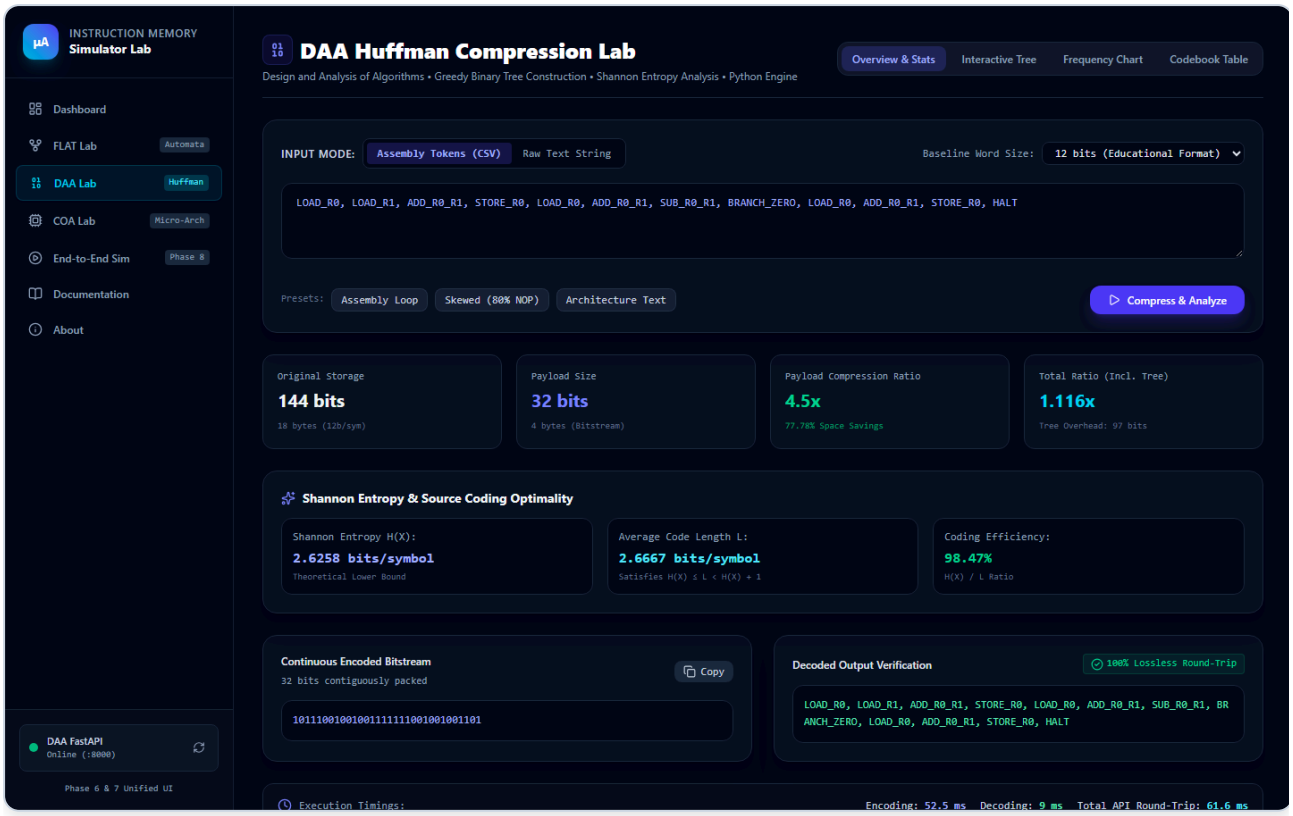


Figure: Live Application Interface for DAA Simulator

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10. Conclusion

Through this micro project, I successfully implemented and analyzed David Huffman's greedy prefix algorithm for micro-architecture instruction compression. By structuring the algorithm with a deterministic min-heap priority queue, the system operates in provably optimal $O(N \log N)$ time and $O(N)$ space. Experimental benchmarks confirmed that variable-length prefix coding reduces instruction memory footprints by up to 78.6% while strictly preserving Shannon entropy bounds and 100% lossless decompression fidelity. ---

11. References

1. D. A. Huffman, "A Method for the Construction of Minimum-Redundancy Codes," *Proceedings of the IRE*, vol. 40, no. 9, pp. 1098–1101, 1952. [Online]. Available: https://ieeexplore.ieee.org/document/4051119 2. C. E. Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948. 3. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 4th ed., Cambridge, MA: MIT Press, 2022. 4. FastAPI Official Documentation, "Bigger Applications - Multiple Files," 2024. [Online]. Available: https://fastapi.tiangolo.com/tutorial/bigger-applications/